

OKLAHOMA CITY ROGERS, OK, USA

WMO: 723530

Lat: 35.389N

Lon: 97.601W

Elev: 392

StdP: 96.71

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13967

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.7	-7.2	-16.8	0.9	-7.1	-14.3	1.1	-4.4	14.4	4.7	12.9	6.2	5.3	0	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	38.1	23.3	36.5	23.4	34.9	23.5	25.6	32.9	25.1	32.4	24.5	31.8	5.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	19.3	28.9	23.0	18.6	28.2	22.4	18.0	27.7	80.9	32.9	78.2	32.4	75.9	31.8	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.5	11.3	10.3	-13.7	39.8	2.6	2.1	-15.6	41.4	-17.2	42.6	-18.7	43.8	-20.6	45.3
			-14.5	26.6	2.5	0.8	-16.3	27.2	-17.7	27.7	-19.1	28.1	-21.0	28.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.3	4.0	6.2	11.1	15.8	20.7	25.5	28.0	27.7	23.3	16.8	10.4	4.9
	DBStd	9.67	5.68	6.20	5.87	4.61	4.06	2.80	2.61	2.80	3.94	4.74	5.15	5.26
	HDD10.0	629	197	134	56	8	1	0	0	0	0	6	57	171
	HDD18.3	1888	444	340	231	101	23	1	0	0	7	84	241	416
	CDD10.0	2917	12	28	91	182	333	464	559	547	400	217	68	13
	CDD18.3	1132	0	1	7	24	97	215	301	289	157	38	3	0
	CDH23.3	12420	2	13	72	227	865	2247	3698	3400	1528	337	30	1
CDH26.7	6028	0	3	13	51	289	1031	2033	1836	678	92	3	0	
Wind	WSAvg	5.0	5.2	5.4	5.9	6.1	5.3	4.9	4.3	4.1	4.3	4.9	5.2	5.0
Precipitation	PrecAvg	878	29	38	66	80	135	111	75	75	94	83	50	39
	PrecMax	1485	94	118	204	192	495	372	250	253	301	335	204	207
	PrecMin	459	0	0	1	4	22	14	0	0	15	9	0	3
	PrecStd	208	24	28	46	45	96	73	56	53	64	63	47	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.7	25.1	28.4	30.9	34.4	37.4	40.3	40.4	38.3	31.7	26.7	21.4
		MCWB	11.7	14.1	16.8	18.5	22.2	23.5	22.8	22.9	22.2	20.4	17.4	14.0
	2%	DB	18.8	21.8	25.3	28.0	31.7	35.1	38.3	38.0	34.6	29.2	23.6	18.6
		MCWB	11.2	12.8	15.6	18.1	22.1	23.6	23.4	23.2	22.7	19.8	15.7	13.0
	5%	DB	16.0	18.7	22.9	26.0	29.9	33.4	36.6	36.4	32.6	27.1	21.1	15.8
		MCWB	9.7	12.1	14.9	17.3	21.3	23.4	23.7	23.4	22.3	18.8	14.9	10.5
	10%	DB	13.2	16.1	20.5	23.9	27.9	31.9	34.9	34.6	30.6	24.9	18.9	13.4
		MCWB	8.0	10.7	13.8	16.7	20.8	23.3	23.8	23.4	21.7	18.0	13.6	9.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.5	17.1	19.3	21.9	24.6	26.3	26.4	26.1	25.0	23.0	19.7	17.2
		MCDB	18.2	20.8	24.6	26.6	30.5	33.2	34.0	33.6	31.6	28.1	23.0	19.8
	2%	WB	12.8	15.3	17.9	20.3	23.5	25.3	25.6	25.3	24.0	21.8	17.6	14.1
		MCDB	16.5	18.7	22.9	24.9	29.2	31.7	33.3	32.9	31.0	26.3	21.5	16.5
	5%	WB	10.3	13.4	16.7	19.4	22.7	24.6	25.1	24.7	23.4	20.7	16.3	11.6
		MCDB	14.7	17.5	21.2	23.7	28.0	31.1	32.8	32.4	29.8	24.5	19.6	14.7
	10%	WB	8.1	10.9	15.0	18.1	21.7	23.9	24.5	24.1	22.7	19.5	14.7	9.0
		MCDB	12.8	16.0	19.4	22.3	26.6	30.2	32.1	31.8	28.7	23.4	18.1	12.9
Mean Daily Temperature Range	5% DB	MDBR	11.6	11.8	12.1	12.1	11.0	11.0	12.0	12.0	11.6	11.9	11.9	10.9
		MCDBR	16.3	16.3	15.2	14.6	12.9	12.8	14.4	14.6	13.9	14.2	14.9	14.9
		MCWBR	9.8	9.5	8.2	7.7	5.9	4.7	4.0	4.3	5.0	6.8	8.6	9.4
	5% WB	MCDBR	14.1	13.8	12.3	11.7	11.2	11.0	12.0	11.8	11.5	10.6	12.0	12.4
		MCWBR	9.6	9.5	7.5	7.5	6.0	5.0	4.4	4.3	4.6	6.3	8.6	9.6
Clear-Sky Solar Irradiance	taub	0.293	0.298	0.338	0.376	0.414	0.423	0.424	0.433	0.387	0.345	0.311	0.289	
	taud	2.508	2.509	2.419	2.327	2.294	2.309	2.337	2.256	2.364	2.473	2.504	2.520	
	Ebn at Noon	917	951	933	907	873	861	858	844	870	884	886	899	
	Edh at Noon	81	91	109	126	132	130	126	134	114	94	81	76	
All-Sky Solar Radiation	RadAvg	2.79	3.39	4.45	5.50	5.93	6.76	6.83	6.21	5.14	3.91	2.94	2.36	
	RadStd	0.28	0.51	0.28	0.41	0.64	0.64	0.37	0.38	0.45	0.50	0.38	0.26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (38 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page